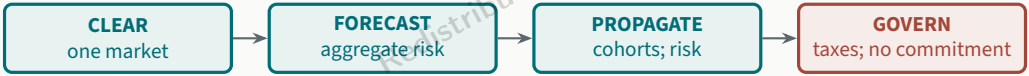
Session 4.4

Optimal Taxation Without Commitment

Put a government into Deck 4.1's economy: a tax rule on the whole distribution, a Markov-perfect fixed point in rules, and the algorithm that finds it

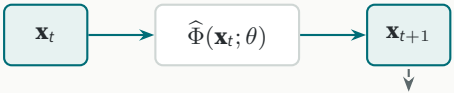


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Zhigang Feng

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Where Decks 4.1 and 4.2 Stopped



Deck 4.1, a household: $\mathbf{x}_{t+1}$ taken as given



this deck, a government: it sees $d\mathbf{x}_{t+1}$

- Deck 4.1: the histogram $\mathbf{x}$ and its matrix $\widehat{\Phi}$, rebuilt at every date inside the graph. A household is atomistic: it takes $\mathbf{x}_{t+1}$ as given.
- Deck 4.2, *Argument 1*: a government that chooses its tax every period needs $d\Gamma'/d\tau$, and only a law of motion in the graph supplies it. Here τ is a function of $\mathbf{x}$, so households respond to it through $\mathbf{x}$.
- This deck puts that government into Deck 4.1's economy.
- **One object is the state, the law of motion and the gradient's path.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Decks 4.1 and 4.2.

The Question: a Tax Nobody Can Promise



- Under commitment the long-run capital tax is zero in the classic models: promise not to tax capital, and people save.
- Capital in place is an inelastic base. A government that chooses afresh each period is tempted to tax it; savers know it.
- Without commitment the problem is a game between today's government and its successors.
- The answer is a rule on the whole distribution, $\tau(\Gamma)$: not a number, not a function of a few moments.
- **Sought: the tax a government chooses when every successor will choose by the same rule.**

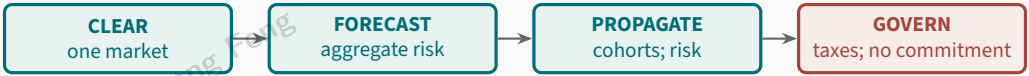
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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Chamley (1986); Judd (1985); Klein, Krusell & Ríos-Rull, *RES* (2008).

Session 4.4

The Economy and Its Equilibrium

Deck 4.1's households, a government that taxes and cannot commit, and a fixed point in rules



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Households

- As in Deck 4.1: ε redrawn every period (i.i.d.), $Q = 5$ Gauss–Hermite nodes (ε_q, p_q), r net of depreciation. A household maximises

$$\sum_{t=0}^{\infty} \beta^t \, U(c_t, l_t, G_t), \qquad U(c, l, G) = u\Big(c - \psi_l \frac{l^{1+\theta_l}}{1+\theta_l}\Big) + \gamma_g \log G, \qquad u(y) = \frac{y^{1-\sigma}}{1-\sigma},$$

  subject to

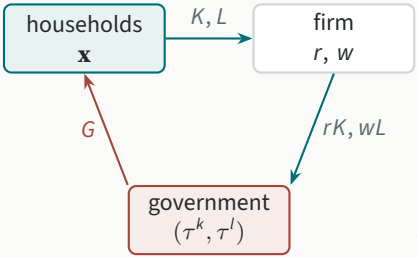
$$c_t + a_{t+1} = \big[1 + (1-\tau_t^k)\, r_t\big]\, a_t + (1-\tau_t^l)\, w_t\, \varepsilon_t\, l_t, \qquad a_{t+1} \geq 0.$$

- τ^k taxes capital income, τ^l labour income; G is a public good, valued with weight γ_g.
- G enters additively, so it changes no household's choice: only the taxes do.
- **Hours still carry no wealth effect: the Frisch elasticity is $1/\theta_l = 3.125$.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. $\beta = 0.95$, $\sigma = 2$, $\theta_l = 0.32$, $\psi_l = 1.5$, $\sigma_\varepsilon = 0.2$.

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Firms, a Balanced Budget, Market Clearing



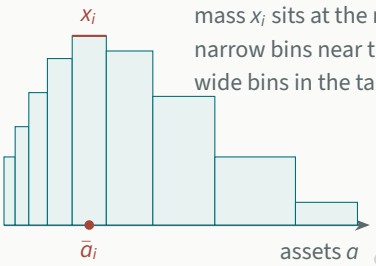
no bonds: G is whatever the taxes raise this period

- Firm: $Y = K^\alpha L^{1-\alpha}$, $r = \alpha(K/L)^{\alpha-1} - \delta$, $w = (1-\alpha)(K/L)^\alpha$.
- Government: picks (τ^k, τ^l) and balances its budget every period, $G = \tau^k rK + \tau^l wL$.
- Markets: $K = \sum_i x_i \bar{a}_i$, $L = \mathbb{E}[\varepsilon\, l(\varepsilon)]$, exactly; the histogram moves by $\mathbf{x}_{t+1} = \mathbf{x}_t \widehat{\Phi}_t$.
- No bonds and no commitment technology: today's government cannot bind its successors.
- G **is a residual, not a choice: two instruments, one budget, every period.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. $\alpha = 0.34$, $\delta = 0.06$.

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The Distribution, Approximated by a Histogram



mass x_i sits at the midpoint $\bar{a}_i$
narrow bins near the limit,
wide bins in the tail

- $N = 140$ bins on $[0, 16]$, midpoints $\bar{a}_1 < \cdots < \bar{a}_N$; the state is the row vector $\mathbf{x} = (x_1, \ldots, x_N)$, $x_i \geq 0$, $\sum_i x_i = 1$.
- Aggregates are sums: $K = \sum_i x_i \bar{a}_i$.
- ε is i.i.d.: $Q = 5$ Gauss–Hermite nodes (ε_q, p_q) integrate it, so $\mathbf{x}$ needs no ε dimension.
- **The government's state is these N numbers.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; as in Deck 4.1. The grid is fixed once, from the classical benchmark.

The Law of Motion $\mathbf{x}_{t+1} = \mathbf{x}_t \widehat{\Phi}_t$, Built Entry by Entry

At date t, for every source bin i and node q

1. saving at the midpoint: $a'_{t,iq} = g_\theta(\varepsilon_q, \bar{a}_i; \mathbf{x}_t)$, kept inside $[\bar{a}_1, \bar{a}_N]$
2. bracket it, $\bar{a}_j \le a'_{t,iq} \le \bar{a}_{j+1}$ with $j = j_{t,iq}$, and weigh the upper neighbour: $\omega_{t,iq} = \dfrac{a'_{t,iq} - \bar{a}_j}{\bar{a}_{j+1} - \bar{a}_j}$
3. send $p_q(1 - \omega_{t,iq})$ of bin i's mass to bin j and $p_q\,\omega_{t,iq}$ to bin $j+1$ (Young's lottery)

$$\widehat{\Phi}_{t,ij'} = \sum_{q=1}^{Q} p_q \Big[(1 - \omega_{t,iq})\, \mathbf{1}\{j' = j_{t,iq}\} + \omega_{t,iq}\, \mathbf{1}\{j' = j_{t,iq} + 1\} \Big], \qquad x_{t+1,j'} = \sum_{i=1}^{N} x_{t,i}\, \widehat{\Phi}_{t,ij'}.$$

- Row i says where bin i's households go: rows sum to one, so mass is conserved.
- Exact in the mean: $\sum_{j'} \widehat{\Phi}_{t,ij'} \bar{a}_{j'} = \sum_q p_q a'_{t,iq}$, so K_{t+1} equals what households save.
- $\widehat{\Phi}_t$ **moves with θ and with $\mathbf{x}_t$, through a': it is rebuilt at every date, and ω is linear in a', so it can be differentiated.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Young, *JEDC* (2010); Deck 4.1. $N \times Q$ network calls per date; no simulation, no sampling noise.

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Household Conditions

- Saving: the Euler condition with the limit, equality whenever $a' > 0$,

$$u'(y_t) \;\geq\; \beta\,\mathbb{E}_{\varepsilon'}\Big\{\big[1+(1-\tau^k_{t+1})\,r_{t+1}\big]\,u'(y_{t+1})\Big\}, \qquad y = c - \psi_l \frac{l^{1+\theta_l}}{1+\theta_l}.$$

- Hours, in closed form:

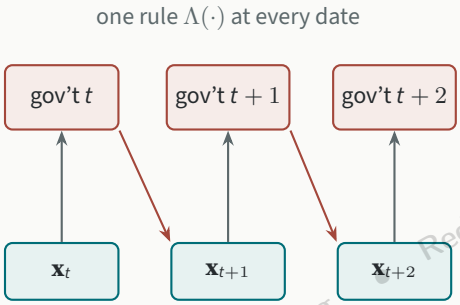
$$(1-\tau^l)\,w\,\varepsilon = \psi_l\, l^{\theta_l} \quad\Longrightarrow\quad l(\varepsilon) = \Big(\frac{(1-\tau^l)\,w\,\varepsilon}{\psi_l}\Big)^{1/\theta_l}.$$

- *Tomorrow's* capital tax sits inside *today's* saving decision: expectations about successors move the economy now.
- Hours contain no a. Given τ^l and K, labour L, both prices and both tax bases are fixed, whatever the shape of $\mathbf{x}$.
- **Under GHH the marginal utility of consumption is $u'(y)$, so $U_c = u'(y)$ in every condition below.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Greenwood, Hercowitz & Huffman, *AER* (1988).

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State, Rule, Timing



- State: $(\varepsilon, a; \mathbf{x})$ for a household; $\mathbf{x}$ alone for the government.
- A **Markov rule** $\Lambda(\mathbf{x}) = \left(\tau^k(\mathbf{x}), \tau^l(\mathbf{x})\right)$, with $G(\mathbf{x})$ from the budget: no calendar time, no history.
- Every government, at every date, uses the same rule, and that rule is the unknown.
- Households take a perceived rule $\tilde{\Lambda}$ and law of motion Φ as given, and best-respond.
- **The Markov restriction, not foresight, is what makes the rule time-consistent.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Klein, Krusell & Ríos-Rull (2008).

The Two Problems

(P-1) Household. Given prices, G, the perceived rule $\tilde{\Lambda}$ and $\mathbf{x}_+ = \mathbf{x}\widehat{\Phi}$,

$$V(\varepsilon, a; \mathbf{x}) = \max_{0 \le a'} \Big\{ U(c, l, G) + \beta \sum_{q'} p_{q'} \, V(\varepsilon_{q'}, a'; \mathbf{x}_+) \Big\}.$$

(P-2) Government (Deck 4.2's V^G). Given the successors' rule $\tilde{\Lambda}$ and its value $\widetilde{W}$,

$$W(\mathbf{x}) = \max_{\tau^k, \tau^l} \Big\{ \sum_{i,q} x_i \, p_q \, U(c_{iq}, l_q, G) + \beta \, \widetilde{W}(\mathbf{x}_+) \Big\},$$

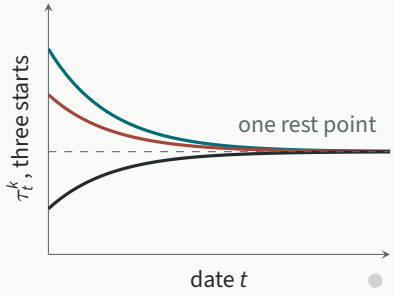
subject to the budget, prices, the household's conditions at *every* node $(\bar{a}_i, \varepsilon_q)$, and $\mathbf{x}_+ = \mathbf{x}\widehat{\Phi}$.

- Welfare is utilitarian: each node counts by its mass $x_i p_q$.
- **The asymmetry: the household takes $\mathbf{x}_+$ as given; the government is not atomistic and sees how its choice moves $\mathbf{x}_+$.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Klein, Krusell & Ríos-Rull (2008).

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Markov-Perfect Equilibrium



- A **Markov-perfect equilibrium**: saving and value functions, prices, a rule Λ with value W, and perceptions, with

  (a) households solving (P-1); (b) $\Lambda(\mathbf{x})$ solving (P-2) at every $\mathbf{x}$; (c) perceived = actual: $\tilde{\Lambda} = \Lambda$, $\widetilde{W} = W$, Φ from households' saving; (d) markets clearing.
- No aggregate shocks: from *any* $\mathbf{x}_0$, taxes and aggregates must settle to constants.
- **A sharp test: paths from different starts must meet.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. Schematic curves: direction and shape only.

Three Computational Challenges

Challenge	Deck 4.2's methods	Here
the distribution is a state	learned moments of a panel; a histogram; a history of shocks	Deck 4.1's histogram, and quadrature for ε
inequality constraints	Fischer–Burmeister: two inequalities become one equation	the same, at every node, priced by a penalty
strategic interaction	none of the three has a government that re-chooses	Deck 4.1's matrix, with the gradient passed through it

- Simulating Γ_+ works when nobody acts on the distribution; with a player who does, it breaks the forward and backward passes (Deck 4.2, *Argument 2*).
- **Deck 4.1 settled the first row. This deck takes the other two, then asks: is the answer right?**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Maliar et al. (2021); Azinovic et al. (2022); Han, Yang & E (2026).

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Implementability: One Equation per Household State

- At every node $(\bar{a}_i, \varepsilon_q)$ the government must respect the Euler condition with the limit, as one equation:

$$\mathrm{FB}(\vartheta_1, \vartheta_2) = \vartheta_1 + \vartheta_2 - \sqrt{\vartheta_1^2 + \vartheta_2^2 + \eta^2} = 0, \qquad \bar{E}_{iq} = \mathrm{FB}_{iq} / u'(y_{iq}),$$

$$\vartheta_1 = u'(y) - \beta \sum_{q'} p_{qq'} \left[1 + (1 - \bar{\tau}^k(\mathbf{x}_+))\, r(\mathbf{x}_+)\right] u'(y_{+,q'}), \qquad \vartheta_2 = a'.$$

- $\mathrm{FB} = 0$ when both are non-negative and one is zero; $\eta = 10^{-4}$. $\bar{E}$ is a relative Euler error.
- a' is the saving *after* the clamp, as the budget and $\widehat{\Phi}$ use it (Deck 4.1: pre-clamp $\tilde{a}$).
- Tomorrow's tax is the *perceived* rule $\bar{\tau}^k$, at the $\mathbf{x}_+$ today's saving produces.
- **The function is held fixed; the state it is evaluated at is not. There the strategic channel lives.**

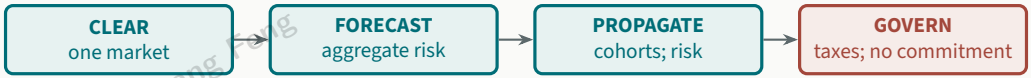
Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Maliar, Maliar & Winant (2021).

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Session 4.4

The Ingredients and the Algorithm

Three networks, one penalised objective, a rollout through the matrix, and a frozen copy of the future



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Three Trained Networks

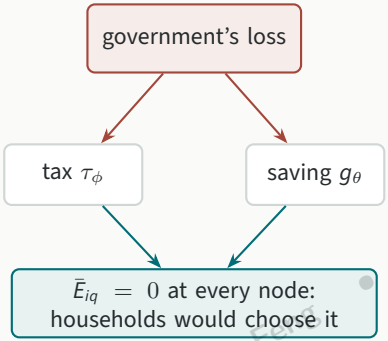
Network	Input	Out	Head
saving g_θ	$(\varepsilon, a, \mathbf{x})$	1	Deck 4.1's change head, $\tilde{a} = a + D(2s-1)$, $a' = \min\{\max\{\tilde{a}, 0\}, \bar{a}\}$
tax rule τ_ϕ	$\mathbf{x}$	2	a sigmoid times a cap: $\tau^k = \tau^k_{\max}\sigma(\cdot)$, $\tau^l = \tau^l_{\max}\sigma(\cdot)$
aggregate value $\bar{V}_\chi$	$\mathbf{x}$	1	linear: the government's value beyond the rollout
household critic V_ψ	$(\varepsilon, a, \mathbf{x})$	1	Deck 4.1's critic: loaded, not trained here

- Each: two hidden layers of 256 rectified-linear units. The tax rule reads the whole histogram, so it can respond to its *shape*.
- **The saving network reads $\mathbf{x}$, not the tax: in equilibrium the tax is itself a function of $\mathbf{x}$.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. $D = 1.3$, as in Deck 4.1.

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The Government Chooses the Allocation Too

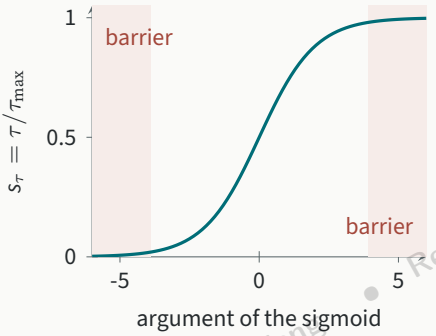


- Deck 4.1 trained saving on the household's own Bellman equation. Here it learns only from the government's loss.
- The government picks the tax *and* the saving rule; $\bar{E} = 0$ makes it one households would choose.
- The **primal approach** of Ramsey taxation, with a Markov government in place of a committed one.
- **The rule is the households' only as far as the penalty drives $\bar{E}$ to zero.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Chari & Kehoe, *Handbook of Macroeconomics* (1999), ch. 26.

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The Tax Rule's Head Decides What Can Be Learned

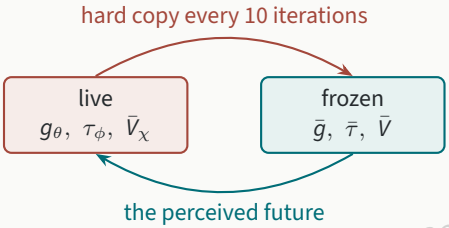


- $\tau^k(\mathbf{x}) = \tau^k_{\max} s_\tau$, $s_\tau = \sigma(\ell(\mathbf{x}))$, ℓ the last layer: its slope carries $\tau^k_{\max} s_\tau(1 - s_\tau)$.
- At the cap, $s_\tau \to 1$ and the gradient vanishes. A constant tax is a trivial fixed point: the run looks converged.
- Remedy one: a cap $\tau_{\max}$ wide enough for an interior optimum. A solver setting, not economics.
- Remedy two: a hinge barrier, zero inside, $B_\sigma = \sum \left[(0.02 - s_\tau)^2_+ + (s_\tau - 0.98)^2_+\right]$.
- **A floor under the gradient wherever τ lands, or the rule cannot learn to respond to $\mathbf{x}$.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

Perceived Equals Actual: the Frozen Copy



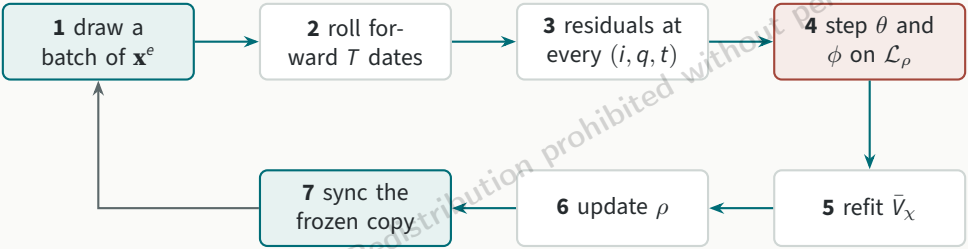
today's choices are trained; tomorrow's rule, saving and value are read from the copy

- The frozen copy supplies $\bar{\tau}^k(\mathbf{x}_+)$ and $\bar{g}$ inside ϑ_1, and $\bar{V}(\mathbf{x}_T)$ at the end of the rollout.
- Inside one step these perceptions do not move: they are targets, not parameters.
- Copying the live networks into the frozen ones imposes consistency: perceived $=$ actual.
- It is Session 3's target network, now used to reach a fixed point in rules.
- **At convergence the rule the government expects of its successors is the rule it plays.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Session 3, *Bridge (3): the Household's Actor–Critic*.

The Loop in One Picture

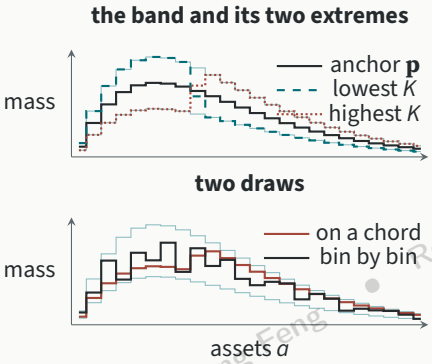


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- Steps 1–4 run four times per iteration, each on a fresh batch; steps 5–7 once, or at longer intervals.
- Steps 2–4 are one forward and one backward pass through $T = 30$ matrices.
- **No outer loop: the fixed point in rules is reached by training and synchronising.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

Step 1a: Sampling Distributions From a Band

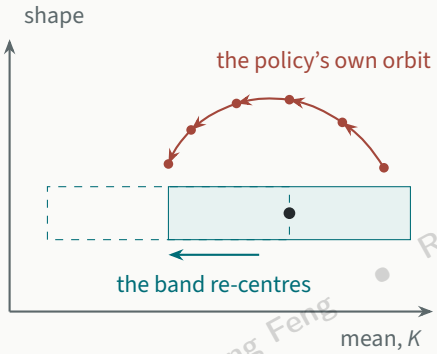


- Around an anchor $\mathbf{p}$, each bin may move by a share ν, $p_i(1-\nu) \le x_i \le p_i(1+\nu)$ (thin lines); the total stays one.
- **Chords**: free mass to the richest, or poorest, bins gives the highest, or lowest, K. Draw K uniformly on the line from $\mathbf{p}$ to that extreme.
- **Bin by bin**: each x_i uniform in its interval, projected back to mass one: every shape, little range in K.
- **Chords give reach in K, bin-by-bin draws give shape: the rule learns only where it is sampled.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. Schematic: an illustrative anchor on 24 bins, $\nu = 0.35$.

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Step 1b: The Band Follows, and the Policy's Own Orbit



- $\mathbf{x}^e, e = 1, \ldots, E$: a batch of starting histograms, one rollout each. Half come from the band (Step 1a).
- Anchor $p_i \propto x_i^* e^{\kappa \bar{a}_i}$, moved every 10 iterations halfway to the classical K^* at current taxes: a taxing government moves both.
- Half from an **orbit**: every 20 iterations, $\mathbf{x} \leftarrow \mathbf{x}\widehat{\Phi}(\mathbf{x}; \theta)$ for 60 steps, no gradient, plus the rule's own stationary $\mathbf{x}$.
- **The band covers off the path; the orbit is accurate on it.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

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Step 2: Roll Forward T Dates

From one sampled $\mathbf{x}_0 = \mathbf{x}^e$, for $t = 0, \ldots, T-1$

1. $K_t = \sum_i x_{t,i} \bar{a}_i$; taxes $(\tau_t^k, \tau_t^l) = \tau_\phi(\mathbf{x}_t)$; hours l_q, then L_t, r_t, w_t: no price loop
2. saving $a'_{t,iq} = g_\theta(\varepsilon_q, \bar{a}_i; \mathbf{x}_t)$; consumption $c_{t,iq}$ from the budget; $y_{t,iq}$
3. revenue $G_t = \sum_{i,q} x_{t,i} p_q \left[\tau_t^k r_t \bar{a}_i + \tau_t^l w_t \varepsilon_q l_q\right]$
4. flow welfare $\bar{U}_t = \sum_{i,q} x_{t,i}\, p_q \left[u(y_{t,iq}) + \gamma_g \log(G_t + 10^{-3})\right]$
5. build $\widehat{\Phi}_t$ from the $a'_{t,iq}$ by Deck 4.1's lottery; $\mathbf{x}_{t+1} = \mathbf{x}_t \widehat{\Phi}_t$, *kept in the graph*

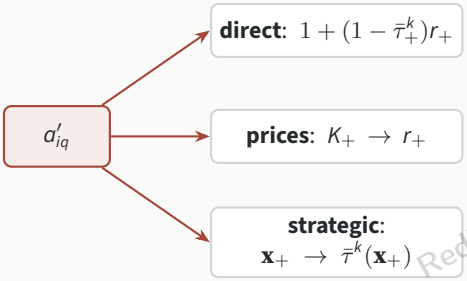
$$W(\mathbf{x}^e; \theta, \phi) = \sum_{t=0}^{T-1} \beta^t \bar{U}_t + \beta^T \bar{V}(\mathbf{x}_T), \qquad \bar{V} \text{ the frozen copy, a constant.}$$

- **Deck 4.1's rollout, with a tax at every date, the public good in welfare, and the matrix left in the graph.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. $T = 30$; utility floored far below any value reached, to guard the logarithm and the power.

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Step 3: The Constraint at Every (i, q, t), and Its Three Channels



each moves ϑ_1; all three stay in the graph

- At each (t, i, q) and batch member: $\bar{E}_{t,iq} = \mathrm{FB}(\vartheta_1, \vartheta_2)/u'(y_{t,iq})$.
- Inside ϑ_1: $\mathbf{x}_+$ from the *live* saving rule; $\bar{\tau}$ and $\bar{g}$ from the frozen copy at $\mathbf{x}_+$.
- Deck 4.1 held a household's continuation fixed: it is its own marginal value (the envelope theorem).
- Here ϑ_1 is a *constraint*: no envelope theorem, and dropping a channel stalls the penalty.
- **The strategic channel, $\mathbf{x}_+ \to \bar{\tau}^k(\mathbf{x}_+)$, is the one no household sees.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

Step 4: The Government's Loss

$$\mathcal{L}_\rho(\theta,\phi) = -W + \sum_{t,i,q} \beta^t x_{t,i} p_q \left[\mu_{t,iq}\bar{E}_{t,iq} + \frac{\rho}{2}\bar{E}^2_{t,iq}\right] + B_{\text{feas}} + B_\sigma(\mathbf{x}^e) + \kappa_{\text{band}} B_{\text{band}}$$

- An augmented Lagrangian, multipliers μ. **As run, $\mu \equiv 0$: a pure quadratic penalty, weight ρ.**
- Weights $\beta^t x_{t,i} p_q$ price each residual by its mass; B_{feas}, same weights: $\left[(-c)_+ + (-y)_+\right]/10^{-4}$.
- $B_{\text{band}} = \sum_t \beta^t \sum_i \left[(p_i(1-\nu) - x_{t,i})_+ + (x_{t,i} - p_i(1+\nu))_+\right]$, $\kappa_{\text{band}} = 20$: rollouts stay in the trained region.
- $B_\sigma = \sum_\tau \left[(0.02 - s_\tau)_+^2 + (s_\tau - 0.98)_+^2\right]$, $s_\tau = \tau/\tau_{\max}$, at the sampled $\mathbf{x}^e$: zero inside, it keeps the sigmoid's slope alive, so the rule cannot freeze at its cap.
- No gradient through the weights, the $u'(y)$ dividing $\bar{E}$, or $\bar{V}(\mathbf{x}_T)$; one backward pass, one Adam step on θ and one on ϕ.
- **Every term but $-W$ is a solver device; the economics is $-W$ and $\bar{E} = 0$.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Nocedal & Wright, *Numerical Optimization* (2006), ch. 17.

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How a Network Responds to the Distribution

- The N bin masses enter the first layer like any other input. For the tax rule, two hidden ReLU layers:

$$\tau_\phi^k(\mathbf{x}) = \tau_{\max}^k \, \sigma\big(\ell_\phi(\mathbf{x})\big), \qquad \ell_\phi(\mathbf{x}) = w_3^\top \mathrm{ReLU}\big(W_2 \, \mathrm{ReLU}(W_1 \mathbf{x}^\top + b_1) + b_2\big) + b_3.$$

- Its response to the mass in bin i, with D_1, D_2 the diagonal masks of active units:

$$\frac{\partial \tau_\phi^k}{\partial x_i} = \tau_{\max}^k \, \sigma'(\ell_\phi) \, \left[w_3^\top D_2 W_2 D_1 W_1\right]_i.$$

- Mass sums to one, so only moves count: δ from bin j to bin i changes τ^k by $\delta\,(\partial_i \tau^k - \partial_j \tau^k)$. A rise in K is one such direction; a mean-preserving spread is another.
- The saving rule $g_\theta(\varepsilon, \bar{a}_i; \mathbf{x})$ and the value $\bar{V}_\chi(\mathbf{x})$ read $\mathbf{x}$ the same way.
- $\partial\tau/\partial\mathbf{x}$ **is learned, not imposed: it is how the government answers inequality, and it enters every saving update.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. Two hidden layers of 256 units; τ^l has its own output ℓ.

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Updating the Tax Rule, One Date at a Time

$$\nabla_\phi \mathcal{L}_\rho = \sum_{t=0}^{T-1} \left[\frac{\partial \mathcal{L}_\rho}{\partial \tau_t^k} \nabla_\phi \tau_\phi^k(\mathbf{x}_t) + \frac{\partial \mathcal{L}_\rho}{\partial \tau_t^l} \nabla_\phi \tau_\phi^l(\mathbf{x}_t) \right] + \nabla_\phi B_\sigma,$$

$$\frac{\partial \mathcal{L}_\rho}{\partial \tau_t^k} = -\beta^t r_t \Big[\underbrace{\frac{\gamma_g K_t}{G_t + 10^{-3}}}_{\text{value of revenue}} - \underbrace{\sum_{i,q} x_{t,i} p_q \, u'(y_{t,iq}) \, \bar{a}_i}_{\text{cost to households}} \Big] + \rho \sum_{i,q} \beta^t x_{t,i} p_q \, \bar{E}_{t,iq} \, \frac{\partial \bar{E}_{t,iq}}{\partial \tau_t^k}.$$

- g_θ does not read τ, so $\mathbf{x}_t$ does not depend on ϕ: each term sits at one date, at the $\mathbf{x}_t$ reached.
- The bracket is the flow trade-off, node by node; the penalty enters through today's $u'(y)$ in ϑ_1.
- **Adam pushes τ_ϕ toward this condition at every $\mathbf{x}$ visited: the rule learns its response to the distribution only where the sampler and rollouts take it.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. r net; $\partial \bar{E} / \partial \tau_t^k$ through $u'(y_{t,iq})$, the divisor held fixed. τ^l: the same form, plus terms through hours, L, r, w.

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Updating the Saving Rule: Through the Law of Motion

$$\mathbf{x}_{t+1} = \mathbf{x}_t\,\widehat{\Phi}_t, \qquad \widehat{\Phi}_t = \widehat{\Phi}\big(a'_t\big), \qquad a'_{t,iq} = g_\theta(\varepsilon_q, \bar{a}_i; \mathbf{x}_t), \qquad \frac{d\mathbf{x}_0}{d\theta} = 0.$$

$$\frac{d\mathbf{x}_{t+1}}{d\theta} = \underbrace{\frac{d\mathbf{x}_t}{d\theta}\,\widehat{\Phi}_t}_{\text{displaced mass, carried on}} + \underbrace{\mathbf{x}_t\,\frac{\partial \widehat{\Phi}_t}{\partial \theta}}_{\text{where today's mass lands}} + \underbrace{\mathbf{x}_t\,\frac{\partial \widehat{\Phi}_t}{\partial \mathbf{x}_t}\,\frac{d\mathbf{x}_t}{d\theta}}_{\text{the saving rule reads } \mathbf{x}_t}$$

- Product rule on $\mathbf{x}_t\widehat{\Phi}_t$, then chain rule: $\widehat{\Phi}_t$ moves with θ directly and through $\mathbf{x}_t$.

Parameters	**The gradient reaches them through**
saving θ	consumption; the three terms, date after date; the taxes at each $\mathbf{x}_t$; the channels of ϑ_1
tax rule ϕ	the budget; G in welfare; $u'(y)$ in $\bar{E}$. Never $\mathbf{x}_+$: g_θ does not read τ
neither	$\bar{V}(\mathbf{x}_\tau)$: the frozen copy, a constant

- **Through θ, the gradient runs back along every $\widehat{\Phi}_t$; through ϕ, only within the date.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. $\partial\widehat{\Phi}/\partial a'$ is the lottery's slope, $\pm 1/(\bar{a}_{j+1} - \bar{a}_j)$, zero where the clamp binds; the third term contracts over $\mathbf{x}_t$. Autograd runs the same chain in reverse.

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Updating the Saving Rule: What the Distribution Sends Back

$$\boldsymbol{\lambda}_t \equiv \frac{d\mathcal{L}_\rho}{d\mathbf{x}_t} = \frac{\partial \ell_t}{\partial \mathbf{x}_t} + \Big(\boldsymbol{\lambda}_{t+1} + \frac{\partial \ell_t}{\partial \mathbf{x}_{t+1}}\Big) J_t, \quad J_t = \frac{\partial \mathbf{x}_{t+1}}{\partial \mathbf{x}_t}, \quad \boldsymbol{\lambda}_T = 0,$$

$$\nabla_\theta \mathcal{L}_\rho = \sum_{t=0}^{T-1} \Big[\frac{\partial \ell_t}{\partial \theta} + \Big(\boldsymbol{\lambda}_{t+1} + \frac{\partial \ell_t}{\partial \mathbf{x}_{t+1}}\Big) \mathbf{x}_t \frac{\partial \widehat{\Phi}_t}{\partial \theta}\Big], \qquad \ell_t : \text{ date } t\text{'s terms of } \mathcal{L}_\rho.$$

In $\partial \ell_t / \partial \mathbf{x}_t$	who sits where, $\sum_q p_q U_{iq}$; prices via K_t; **the tax rule's answer** $\partial \tau_\phi / \partial \mathbf{x}_t$; g_θ's input
In $\partial \ell_t / \partial \mathbf{x}_{t+1}$	in $\bar{E}_t$: $\bar{\tau}^k(\mathbf{x}_{t+1})$, $r(\mathbf{x}_{t+1})$, $\bar{g}(\cdot; \mathbf{x}_{t+1})$: the strategic channel

- The tax rule's answer reaches θ twice: through $\tau_\phi(\mathbf{x}_t)$ along the path, and through $\bar{\tau}(\mathbf{x}_{t+1})$ in each Euler residual.
- **The saving rule is updated knowing how the tax will answer the distribution its own saving creates.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. The previous page run in reverse: one backward pass, $d\mathbf{x}_t / d\theta$ never formed. Penalty weights held fixed: "who sits where" enters through welfare only; $\bar{V}(\mathbf{x}_T)$ sends nothing back.

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What Is New Here

	the state	who acts on $\mathbf{x}$	the gradient through $\mathbf{x}$'s motion
Krusell–Smith; Deck 4.2	mean, moments, histogram, shocks	nobody: households take it as given	one period, or a household panel
Markov taxes, one agent	K	a government	no distribution
HA taxes, commitment	a given $\mathbf{x}_0$	a government, once	a path of taxes, not a rule
this deck	**the whole histogram**	**a government, every date**	**T dates, with $\partial\tau/\partial\mathbf{x}$ inside**

- Without commitment the tax is a function $\tau(\mathbf{x})$; with commitment, a path chosen once.
- **To the authors' knowledge, the first time-consistent tax rule computed on the whole distribution as the state.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. One agent: Klein, Krusell & Ríos-Rull (2008). Commitment: Bhandari et al. (2021); Dyrda & Pedroni (2022).

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Steps 5 and 7: Refit the Value Beyond the Horizon; Sync

- **Targets**, every 20 iterations, at fresh band draws $\mathbf{x}$. As run, flow welfare held forever:

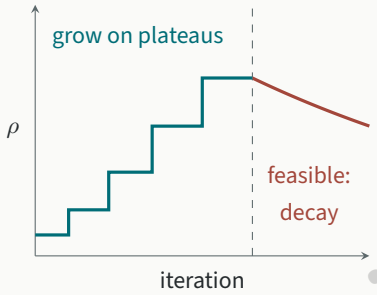
$$\hat{V}(\mathbf{x}) = \frac{\bar{U}(\mathbf{x})}{1-\beta}, \qquad \bar{U}(\mathbf{x}) = \sum_{i,q} x_i \, p_q \, U_{iq}.$$

- **Fit**, every iteration: 20 Adam steps on $\text{mean}[\bar{V}_\chi(\mathbf{x}) - \hat{V}(\mathbf{x})]^2$, batches of 256, after a long prefit.
- **Sync**, every 10 iterations: $g_\theta, \tau_\phi, \bar{V}_\chi$ copied into $\bar{g}, \bar{\tau}, \bar{V}$. A hard copy, not a moving average.
- Beyond T the economy is valued as if it stayed where the rollout ended; at $\beta^T \approx 0.2$ that term is not negligible.
- $\bar{V}$ **supplies a level at the horizon, never a gradient; the rollout supplies the gradient.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

Step 6: The Penalty Weight Follows the Residual



- Each iteration, read $\bar{E}^{\text{test}}$: the mass-weighted mean of $|\bar{E}|$ on a fixed test batch.
- Grow, $\rho \leftarrow \min(1.015\rho, 10^4)$, if the loss stalls ($|\Delta\mathcal{L}/\mathcal{L}| \leq 1\%$) while $\bar{E}^{\text{test}} \geq$ tol and not improving.
- Decay, $\rho \leftarrow 0.99\rho$, once 5 of the last 10 reads are below tol.
- Learning rates scaled by $1/\sqrt{1+0.1\rho}$.
- **Measure tol at the solution: below what is reachable, ρ grows without end.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. Schematic curves: direction and shape only.

The Algorithm

Taxation without commitment, as a trained fixed point

1 warm-start g_θ from the no-tax solve; tax rule near zero; prefit $\bar{V}_\chi$; copy all to the frozen set
2 four times: draw $\mathbf{x}^e$, half band, half orbit (step 1)
3 roll forward T dates, $\widehat{\Phi}_t$ in the graph; form W and every $\widehat{E}_{t,iq}$ (steps 2–3)
4 one backward pass of $\mathcal{L}_\rho$; one Adam step on θ, one on ϕ (step 4)
5 refit $\bar{V}_\chi$; every 20 iterations new targets and orbit, every 10 a moved anchor (step 5)
6 update ρ and the learning rates from the test-batch residual (step 6)
7 every 10 iterations: frozen $\leftarrow$ live, "perceived $=$ actual" (step 7)
8 stop at the iteration budget, or early if the Euler error blows up; keep the best checkpoint

- **Each line is Deck 4.1's machinery or Session 3's target network. What is new is who owns the loss.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

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What the Gradient Requires

Requirement	How it is met
a differentiable transition	lottery weights piecewise linear in a' (Deck 4.1); kinks of measure zero
no sampling noise	quadrature over ε, every asset node: a deterministic loss
a gradient in the tax rule	the barrier keeps the sigmoid off 0 and 1
a gradient at the limit	$\vartheta_2 = a'$ and the smoothed root keep constrained nodes live
memory for the graph	T dates of $N \times Q$ evaluations, $N \times N$ matrices, next-date copies in ϑ_1

- T is a budget. Beyond it, $\bar{V}$ adds a level and no gradient.
- Lab block B: derivatives through up to twenty lottery steps match finite differences to 2×10^{-10}.
- **Every row is a place where a gradient can vanish without an error message.**

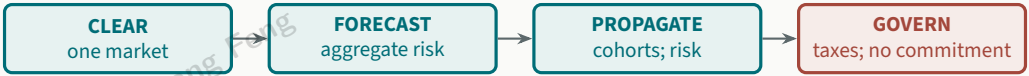
Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; the lab, block B.

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Session 4.4

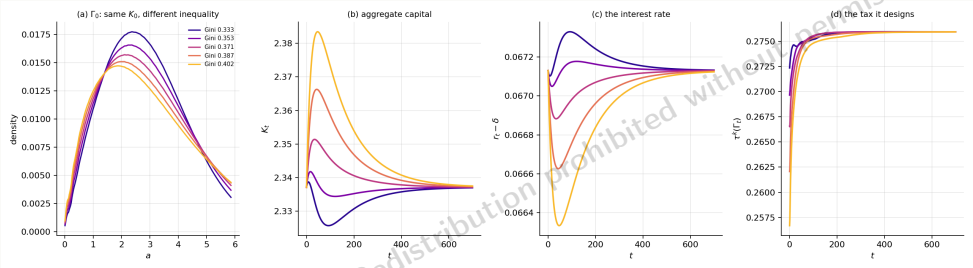
What the Rule Does Along a Path

Start the trained economy from different distributions and let the tax rule, saving and prices run



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At the Same Capital, Inequality Alone Moves the Tax



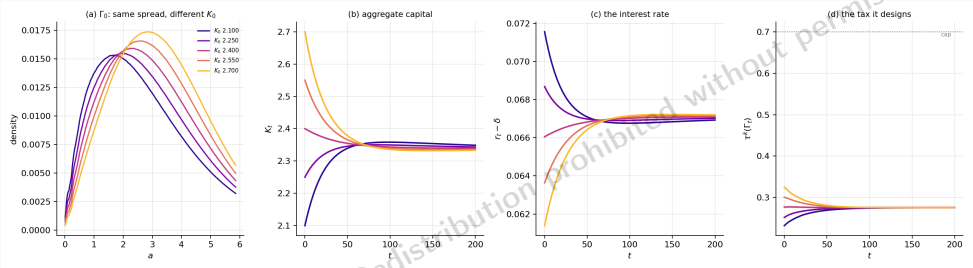
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- Five starts Γ_0 share one K_0 and differ only in inequality (a). At date 0, hours, prices and both tax bases are identical.
- The more unequal start is taxed less on impact (d); its capital overshoots further before returning (b, c).
- Every start converges to the same Γ^* and the same long-run tax.
- **A property of the trained rule: why it responds to shape is not yet settled.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. Simulated with the trained Stage 2A networks.

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At the Same Inequality, More Capital Means a Bigger Opening Levy



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- The mirror experiment: the spread is held and K_0 moves (a). Now the tax base moves too, and the tax responds far more (d).
- The levy is transitional: capital and the interest rate return (b, c), and every start reaches the same long-run rate.
- A cap that never binds at the rest point can still censor the path: size it for the levy.
- **Without commitment: tax the capital inherited, then settle down.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. Simulated with the trained Stage 2A networks.

Bridge: the Lab, and Session 5

Lab, Part I: the classical benchmarks of Decks 4.1–4.3.

Blocks D, E, F: Krusell–Smith by DeepHAM, by Deep Equilibrium Nets

Blocks G, H: the OLG with a market-clearing layer and a homotopy; the retirement OLG in the sequence space.

This deck: a government that differentiates through Γ.

- Cost: GPU-hours, not seconds. Stage 1B alone took about 2.9 GPU-hours, against 30 CPU-seconds for Stage 1A.
- It buys rules on the whole distribution, and the path from any start in the sampled region.
- Next in the research: aggregate shocks, $\widehat{\Phi}$ conditioned on (z, z'); then persistent ε.
- **The lab trains the Deck 4.2 and 4.3 networks in PyTorch, each graded by a gate fixed before the run.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; the Session 4 lab.